

Custom Chat-Bot Assignment: Premium RAG Intelligence Platform

1. Overview of Project

Detail	Description
Project Name	Premium RAG Intelligence Platform
Role/Theme	Enterprise-Grade Document Analysis & Knowledge Assistant
Core Function	Answers user questions using information retrieved from uploaded documents (PDF, DOCX, TXT).
LLM Backend	Llama 4 Scout 17B (via Groq API)
Frameworks	Python, Gradio (UI), LangChain (RAG Pipeline)
Hugging Face link	https://huggingface.co/spaces/UmarO977/premium-rag-chatbot

Description:
The Premium RAG Intelligence Platform is designed to ingest multiple document types, process them into a semantic knowledge base (Vector Store), and provide accurate, source-backed answers using an advanced LLM. The application is built for reliability, featuring robust file handling and a sophisticated, professional user interface to facilitate complex knowledge retrieval from enterprise documents.

2. Enhancements Added (Feature & UI)

A. Robust RAG Pipeline Implementation

- **Semantic Chunking:** Documents are processed using RecursiveCharacterTextSplitter and embedded using a high-quality Sentence-Transformer model (all-MiniLM-L6-v2) before being stored in a FAISS Vector Store, ensuring high-accuracy retrieval.
- **Advanced File Parsing:** Includes custom logic and conditional imports (pdfplumber, pypdf, docx) to reliably handle PDF and DOCX formats, overriding standard loaders

where text extraction quality is poor.

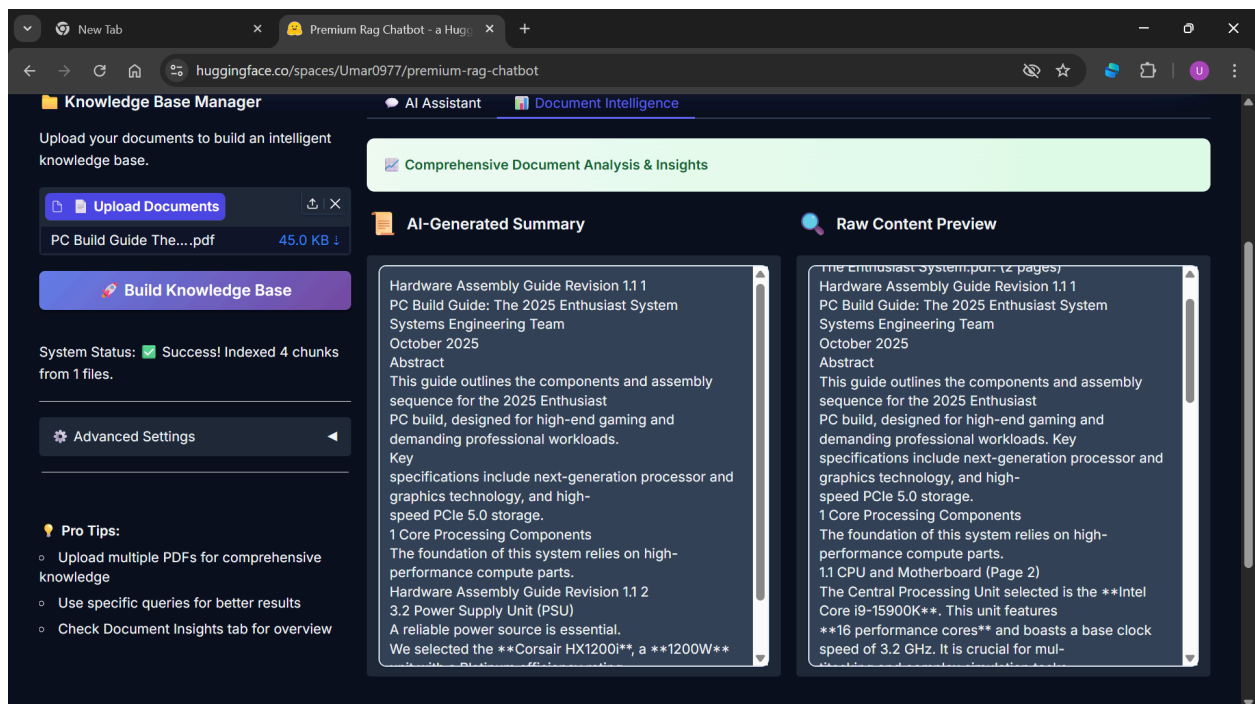
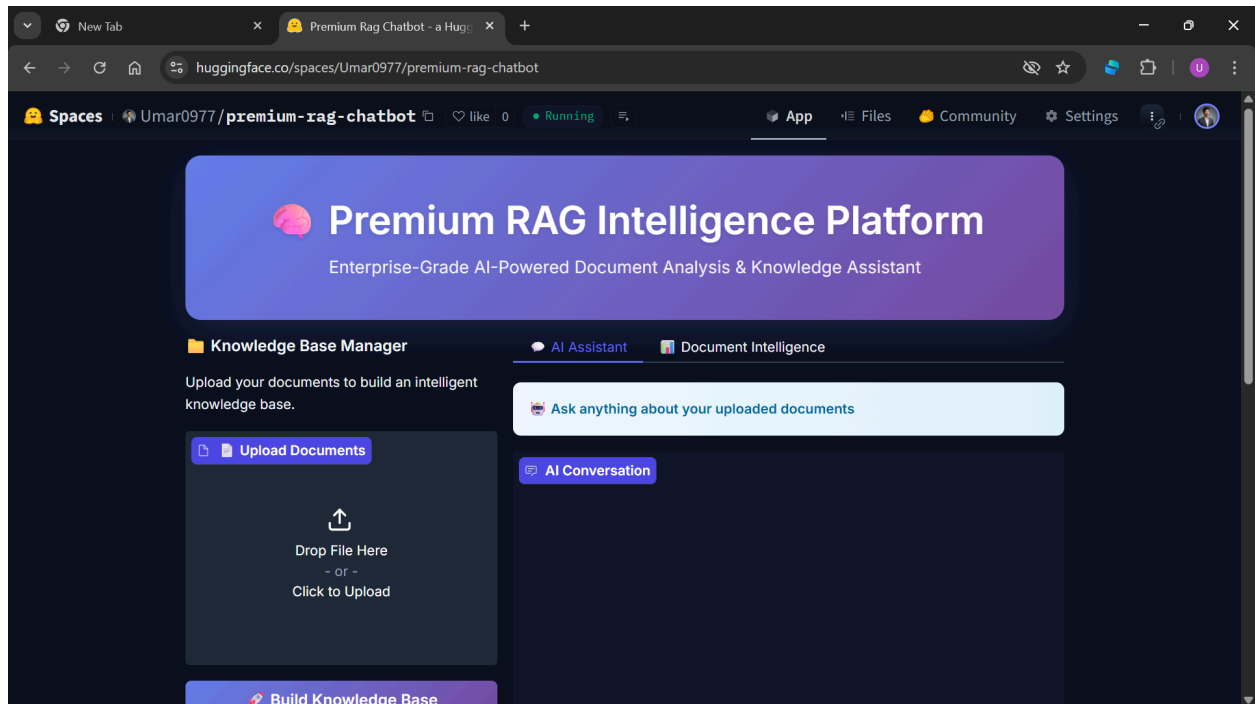
- **Unicode Repair:** Implements `ftfy` (or fallback `regex`) to automatically repair common text encoding errors (`\ufffd` or `mojibake`) that often occur during PDF or DOCX text extraction.
- **Source Citation:** The final response includes clear citations indicating the page number(s) from which the context was retrieved, ensuring transparency and verifiability.

B. Professional Gradio UI/UX

- **Premium Theme:** Utilizes a **Soft** theme with a custom color palette (Indigo/Blue) for a clean, modern, and high-contrast look.
- **Gradient Header:** A custom `gr.HTML` block with gradient background styling is used for a distinguished, branded application header.
- **Document Intelligence Tab:** A separate tab is provided to give users immediate feedback upon processing, displaying an **AI-Generated Summary** (heuristic summary based on the first few paragraphs/list items) and a **Raw Content Preview**.
- **Utility & Actions:** Includes advanced actions like **Query Logging** (to CSV) and a **Download History** button for conversation export.

3. Screenshots of the Running App

Screenshot	Description
3.1: Main RAG Interface	Shows the overall layout: left sidebar for file upload/actions, and the right area defaulted to the "AI Assistant" chat tab. The custom gradient header and premium button styling are visible.
3.2: Document Intelligence Tab	Illustrates the secondary tab, where the AI-Generated Summary and the extracted Raw Content Preview are displayed after successfully processing a document.
3.3: Chat Response with Citation	A screenshot of the chat showing a detailed LLM answer followed by the required source citation (e.g., "Sources: Page(s) 3, 7").
3.4: Advanced Settings	Shows the minimized "Advanced Settings" accordion containing the Enable Logging checkbox and Clear/Export quick actions.



New Tab

Premium Rag Chatbot - a Hug...

huggingface.co/spaces/Umar0977/premium-rag-chatbot

Knowledge Base Manager

Upload your documents to build an intelligent knowledge base.

Upload Documents

PC Build Guide The....pdf45.0 KB

Build Knowledge Base

System Status: Success! Indexed 4 chunks from 1 files.

Advanced Settings

Enable Analytics & Logging

Track queries for audit and Insights

Quick Actions

Clear

Export

AI Assistant

Document Intelligence

Ask anything about your uploaded documents

AI Conversation

specs of cpu?

According to the guide, the specs of the CPU (Intel Core i9-15900K) are:

- 16 performance cores
- Base clock speed: 3.2 GHz

Sources: Page(s) 1, 0

knowledge base.

  Upload Documents



PC Build Guide The....pdf

45.0 KB 



Build Knowledge Base

System Status:  Success! Indexed 4 chunks
from 1 files.

 Advanced Settings 



Enable Analytics & Logging

Track queries for audit and insights



Quick Actions



Clear



Export

4. Challenges Faced

1. **Ensuring Cross-Format Reliability (RAG Pipeline):** The primary challenge was moving beyond a simple PDF loader. Implementing fallback mechanisms for PDF extraction and custom logic for DOCX/TXT files required significantly more complex file path resolution (`_resolve_path`) and error handling to guarantee clean text content for the RAG pipeline.
2. **Maintaining Conversational Context:** The `chat_fn` required careful implementation to extract the previous user/assistant turns from the `history` and inject them back into the LLM prompt (`user_prompt`) alongside the new retrieval context. This prevents the LLM from losing track of the conversation flow.
3. **Complex Gradio Styling:** Achieving the specified "Enterprise-Grade" aesthetic involved extensive custom CSS (`custom_css`) to override Gradio's default theme elements, applying gradients, shadows, and specific font treatments to components like the status message and the main header.